

Digital Design and Computer Organisation Laboratory

UE23CS251A

3rd Semester, Academic Year 2024-25

Date:05/09/2024

Name: Dipak Chaudhari	SRN: PES1UG24CS804	Section K
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Week# 3

Program Number: 1

TITLE: HALF ADDER

WRITE A VERILOG PROGRAM TO MODEL A HALF ADDER. GENERATE THE VVP OUTPUT AND SIMULATION WAVEFORM USING GTKWAVE. VERIFY THE OUTPUT AND WAVEFORM WITH THE AND GATE TRUTH TABLE

Verilog Code Screenshot

Verilog VVP Output Screen Shot

GTKWAVE Screenshot

Output Table to be completed and included

HALF ADDER

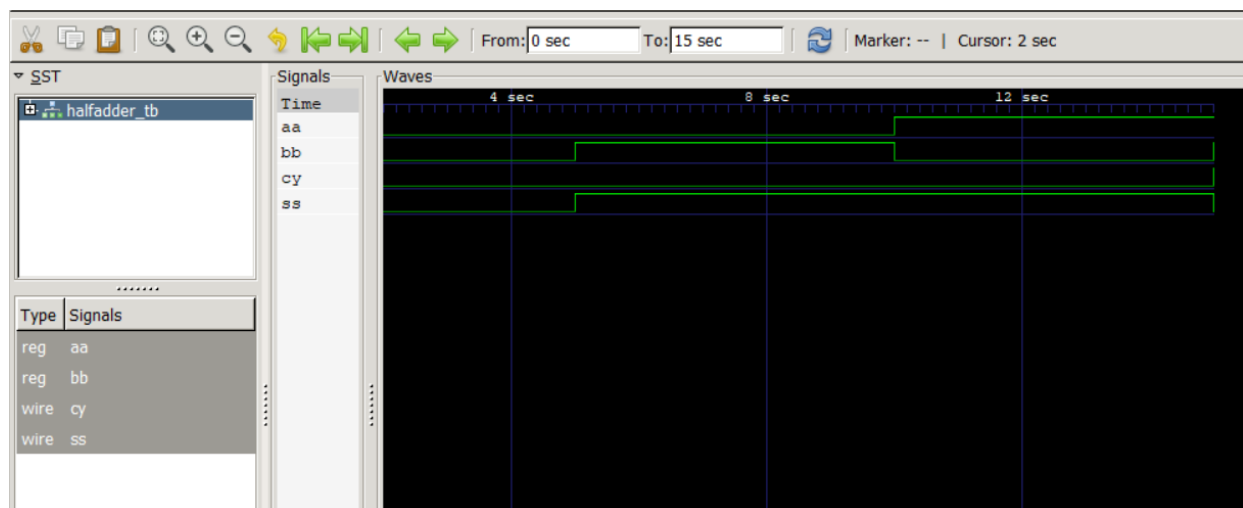
half_adder.v file

```
1  module halfadder (  
2      input wire a, b,  
3      output wire sum, c  
4  );  
5  
6      xor x0 (sum, a, b);  
7      and a0 (c, a, b);  
8  endmodule
```

half_adder_tb.v file

```
module halfadder_tb;  
    reg aa, bb;  
    wire ss, cy;  
    halfadder add1 ( aa, bb, ss, cy );  
    initial begin  
        aa = 1'b0; bb = 1'b0;  
        #5 aa = 1'b0; bb = 1'b1;  
        #5 aa = 1'b1; bb = 1'b0;  
        #5 aa = 1'b1; bb = 1'b1;  
    end  
    initial begin  
        $monitor($time, "a=%b, b=%b,sum=%b,carry=%b", aa, bb, ss, cy);  
    end  
    initial begin  
        $dumpfile("halfadder.vcd");  
        $dumpvars(0, halfadder_tb);  
    end  
endmodule
```

```
C:\iverilog\bin\WEEK-3\WEEK>vvp sampletest
VCD info: dumpfile halfadder.vcd opened for output.
      0a=0, b=0,sum=0,carry=0
      5a=0, b=1,sum=1,carry=0
     10a=1, b=0,sum=1,carry=0
     15a=1, b=1,sum=0,carry=1
```



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Week#____3_____

Program Number: ____2____

TITLE: FULL ADDER

WRITE A VERILOG PROGRAM TO MODEL A FULL ADDER. GENERATE THE VVP OUTPUT AND SIMULATION WAVEFORM USING GTKWAVE. VERIFY THE OUTPUT AND WAVEFORM WITH THE AND GATE TRUTH TABLE

Verilog Code Screenshot

Verilog VVP Output Screen Shot

GTKWAVE Screenshot

Output Table to be completed and included

FULL ADDER

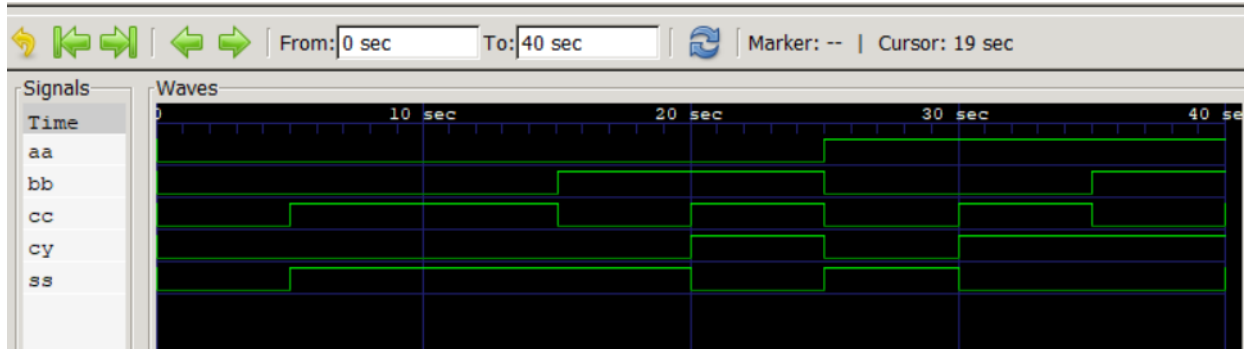
full_adder.v file

```
1  module fulladder (  
2      input wire a, b, cin,  
3      output wire sum, cout  
4  );  
5  
6      wire [4:0] t;  
7  
8      xor x0 (t[0], a, b);  
9      xor x1 (sum, t[0], cin);  
10  
11     and a0 (t[1], a, b);  
12     and a1 (t[2], a, cin);  
13     and a2 (t[3], b, cin);  
14  
15     or o0 (t[4], t[1], t[2]);  
16     or o1 (cout, t[4], t[3]);  
17 endmodule
```

full_adder_tb.v file

```
v  
module fulladder_tb;  
    reg aa, bb, cc;  
    wire ss, cy;  
    fulladder add1 ( .a(aa), .b(bb), .cin(cc), .sum(ss), .cout(cy) );  
    initial begin  
        #0 aa = 1'b0; bb = 1'b0; cc = 1'b0;  
        #5 aa = 1'b0; bb = 1'b0; cc = 1'b1;  
        #5 aa = 1'b0; bb = 1'b0; cc = 1'b1;  
        #5 aa = 1'b0; bb = 1'b1; cc = 1'b0;  
        #5 aa = 1'b0; bb = 1'b1; cc = 1'b1;  
        #5 aa = 1'b1; bb = 1'b0; cc = 1'b0;  
        #5 aa = 1'b1; bb = 1'b0; cc = 1'b1;  
        #5 aa = 1'b1; bb = 1'b1; cc = 1'b0;  
        #5 aa = 1'b1; bb = 1'b1; cc = 1'b1;  
    end  
    initial begin  
        $monitor($time, "a=%b, b= %b, c= %b,sum= %b,carry= %b", aa, bb, cc, ss, cy);  
    end  
    initial begin  
        $dumpfile("fulladder.vcd");  
        $dumpvars(0, fulladder_tb);  
    end  
endmodule
```

```
0a=0, b= 0, c= 0, sum= 0, carry= 0
5a=0, b= 0, c= 1, sum= 1, carry= 0
15a=0, b= 1, c= 0, sum= 1, carry= 0
20a=0, b= 1, c= 1, sum= 0, carry= 1
25a=1, b= 0, c= 0, sum= 1, carry= 0
30a=1, b= 0, c= 1, sum= 0, carry= 1
35a=1, b= 1, c= 0, sum= 0, carry= 1
40a=1, b= 1, c= 1, sum= 1, carry= 1
```



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Week# 3

Program Number: 3

TITLE: RIPPLE Adder

WRITE A VERILOG PROGRAM TO MODEL A RIPPLE ADDER. GENERATE THE VVP OUTPUT AND SIMULATION WAVEFORM USING GTKWAVE. VERIFY THE OUTPUT AND WAVEFORM WITH THE AND GATE TRUTH TABLE

Verilog Code Screenshot

Verilog VVP Output Screen Shot

GTKWAVE Screenshot

Output Table to be completed and included

RIPPLE ADDER

```
`timescale 1 ns / 100 ps
`define TESTVECS 10

module tb;
  reg clk, reset;
  reg [3:0] i0, i1;
  reg cin;
  wire [3:0] o;
  wire cout;
  reg [8:0] test_vecs[0:(`TESTVECS-1)];
  integer i;
  initial begin
    $dumpfile("4bitrippleadder.vcd");
    $dumpvars(0, tb);
  end
  initial begin
    reset = 1'b1;
    #12.5 reset = 1'b0;
  end
  initial clk = 1'b0;
  always #5 clk = ~clk;
  initial begin

    test_vecs[0] = 9'b000000000;
    test_vecs[1] = 9'b000000001;
    test_vecs[2] = 9'b000100010;
    test_vecs[3] = 9'b000100011;
    test_vecs[4] = 9'b001000100;
    test_vecs[5] = 9'b101010110;
    test_vecs[6] = 9'b101010111;
    test_vecs[7] = 9'b110011010;
    test_vecs[8] = 9'b111111100;
    test_vecs[9] = 9'b111111101;
  end
  initial {i0, i1, cin, i} = 0;
  fulladder u0 ( i0, i1, cin, o, cout );
  initial begin
    #6
    for (i = 0; i < `TESTVECS; i = i + 1) begin
      #10;
      {i0, i1, cin} = test_vecs[i];
    end
    #100 $finish;
  end
  always @(i0 or i1 or cin)
  $monitor( "At time = %t, i0=%b, i1=%b, cin=%b, Sum=%b, Carry=%b", $time, i0, i1, cin, o, cout );
endmodule
```

ripple_adder

```
module fulladd(input wire a,b,cin,output wire sum,cout);
  wire [4:0] t;
  xor x0 (t[0],a,b);
  xor x1 (sum,t[0],cin);
  and a0 (t[1],a,b);
  and a1 (t[2],a,cin);
  and a2 (t[3],b,cin);
  or o0 (t[4],t[1],t[2]);
  or o1 (cout,t[4],t[3]);
endmodule

module fulladdR (
  input wire [3:0] a, b,
  input wire cin,
  output wire [3:0] sum,
  output wire cout
);

  wire [2:0] c;

  fulladd u0 (
    a[0], b[0], cin, sum[0], cout0
  );

  fulladd u1 (
    a[1], b[1], cout0, sum[1], cout1
  );

  fulladd u2 (
    a[2], b[2], cout1, sum[2], cout2
  );

  fulladd u3 (
    a[3], b[3], cout2, sum[3], cout
  );

endmodule
```

```
At time = 0, i0=0000, i1=0000, cin=0, Sum=0000, Carry=0
At time = 260, i0=0000, i1=0000, cin=1, Sum=0001, Carry=0
At time = 360, i0=0001, i1=0001, cin=0, Sum=0010, Carry=0
At time = 460, i0=0001, i1=0001, cin=1, Sum=0011, Carry=0
At time = 560, i0=0010, i1=0010, cin=0, Sum=0100, Carry=0
At time = 660, i0=1010, i1=1011, cin=0, Sum=0101, Carry=1
At time = 760, i0=1010, i1=1011, cin=1, Sum=0110, Carry=1
At time = 860, i0=1100, i1=1101, cin=0, Sum=1001, Carry=1
At time = 960, i0=1111, i1=1110, cin=0, Sum=1101, Carry=1
At time = 1060, i0=1111, i1=1110, cin=1, Sum=1110, Carry=1
```

